

Sharif Uddin

Electrical Engineering | RF, Microwave & Antenna Research

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[Google Scholar](#) | [Website](#) | [LinkedIn](#)

Research Interests

RF and Microwave Engineering; Antenna Design and Characterization; Circularly Polarized Antennas; Full-Duplex Wireless Systems; MIMO Antennas and Mutual Coupling Reduction; mmWave and Terahertz Antennas; Antennas-on-chip; RF Integrated Circuit, Antenna Fabrication and Measurement.

Education

Master of Science in Electrical Engineering [Fall 2025–Present]
Washington State University Vancouver, USA GPA: [3.93/4.00]

- Year of completion: May 2027
- Thesis: Wideband circularly polarized antennas for full-duplex wireless communication and relay application.
- Advisor: Dr. Tutku Karacolak.

Master of Science in Electrical, Electronic and Communication Engineering – Only Coursework Completed [2022]–[2024]
Military Institute of Science and Technology (MIST), Bangladesh CGPA: 3.58/4.00

- Coursework completed; degree not awarded

B.Sc. in Electrical & Electronic Engineering [2016]–[2021]
Rajshahi University of Engineering & Technology (RUET), Bangladesh CGPA: 3.54/4.00

Research Experience

Graduate Researcher Fall 2025–Present
Washington State University, USA

- Design and investigate **differentially fed wideband circularly polarized antennas** for full-duplex & wireless relay communication systems.
- Perform full-wave electromagnetic simulations using **ANSYS HFSS**, followed by **antenna fabrication** and experimental validation.
- Characterize fabricated antennas in **anechoic chamber** and microwave measurement using **vector network analyzer**.

Selected Research Projects

Differentially Fed Wideband Circularly Polarized Antenna for Full-Duplex Communication

- Investigating wideband CP antenna architectures for simultaneous transmit and receive operation.
- Studying isolation mechanisms including polarization, spatial/radiation-pattern, structural, and decoupling-based techniques.
- Experimental validation through fabricated prototypes and measured differential S-parameters through a hybrid coupler, radiation characteristics, gain, efficiency, MIMO characterization: ECC, DG, and TARC.

Antenna Design, Simulation, Fabrication, and Measurement of Circularly Polarized Relay Antenna

- Designed and optimized wideband and circularly polarized relay antennas using full-wave electromagnetic simulation.
- Fabricated antenna prototypes and performed experimental characterization using an anechoic chamber.

Publications

Published

- [1] **S. Uddin**, et al., “A Compact Wideband Circularly Polarized MIMO Antenna for High Data Rate Communications,” 2026 *National Radio Science Meeting (NRSN)*, 2026.
- [2] **S. Uddin**, et al., “An Improved Switching Control Technique of a Three-Phase NPC Multilevel Converter in Reducing the Neutral Point Potential Swing,” *IEEE Xplore*, 2024.
- [3] A. Hossain **S. Uddin**, et al., “Wavelet and Spectral Analysis of Normal and Abnormal Heart Sound for Diagnosing Cardiac Disorders,” *BioMed Research International*, 2022.

Manuscripts Under Review

- [1] **S. Uddin**, et al., “Differentially Fed Wideband Circularly Polarized Antenna for Full-Duplex Communications,” manuscript submitted for publication, 2026.

Teaching Experience

Graduate Teaching Assistant, Department of Electrical & Computer Engineering

[2025]–[Present]

Washington State University Vancouver, USA

- Grading and providing feedback on homework assignments and lab reports, as well as facilitating laboratory sessions for courses including • Antenna Design and Analysis • RF Devices and Circuits • Electromagnetic Fields and Waves • Principles of Solid-State Devices.

Lecturer, Department of Electrical & Electronic Engineering

[2023]–[2025]

Northern University Bangladesh, Bangladesh

- Taught undergraduate courses in Electrical and Electronic Engineering and supervised undergraduate capstone projects, including poster presentations.

Lecturer, Department of Electrical & Electronic Engineering

[2021]–[2023]

Prime University, Bangladesh

- Contributed to undergraduate teaching, laboratory instruction, student mentoring, and academic and extracurricular activities, including serving as sports advisor.

Technical Skills

Electromagnetics	Antenna design on rigid substrate, RF/microwave engineering, Circularly polarized antennas, MIMO, full-duplex systems, mmWave/THz, Antennas-on-chip and RF integrated circuit concepts
Simulation	ANSYS HFSS, CST Studio Suite, Advance Design System (ADS)
Measurement	Using Anechoic Chamber, Keysight FieldFox N9952A VNA
Fabrication	LPKF laser & electronics milling machine, Microfabrication: Thermal Oxidation, Photolithography, Sputtering, Lift-off, Etching, and Thin-film de-position
Programming & Data Analysis	MATLAB, Python (Google Colab), Machine Learning, C/C++, Design-Expert: Design of Experiments (DoE), ANOVA

Professional Activities

- Presented research at the 2026 National Radio Science Meeting (NRSF), Boulder, Colorado, USA.
- Presented research at the 6th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT), 2024.

Additional Information

English Proficiency: IELTS Overall 7.0 (L 8.0, R 7.0, W 7.0, S 6.5)

GRE: Quantitative 159, Verbal 144, AWA 3.5

References

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